

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 20



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Recap of Previous Lecture



Topic

Identical function



Topic

Function composition



Topics to be Covered



Topic

Algebraic structure

Topic

Semi-group

Topic

Monoid

Topic

Group

Topic

Abelian group / Commutative group

Topic

Practice questions



Topic : Special Sets



$\mathbb{N}$ = Set of all natural numbers

$\mathbb{Z}$ = Set of all integers

$\mathbb{Q}$ = Set of all rational numbers

$\mathbb{Q}^*$ = Set of all non-zero rational numbers

$\mathbb{R}$ = Set of all real numbers.



Topic : Algebraic Structure

A non-empty set 'S' w.r.t. binary operation '*' is called an algebraic structure (groupoid/magma) if and only if

$$a * b \in S, \quad \forall a, b \in S$$

it is called
closure property

{ i.e. Set 'S' is closed }
w.r.t. binary opⁿ * }



	Algebraic Structure (Groupoid/Magma)	Semi-Group	Monoid	Group	Abelian Group
$(\mathbb{N}, +)$	✓				
$(\mathbb{N}, \cdot)$	✓				
$(\mathbb{N}, -)$	$2 - 3 = -1 \notin \mathbb{N} \times$				
$(\mathbb{N}, \div)$	X				
$(\mathbb{Z}, +)$	✓				
$(\mathbb{Z}, \cdot)$	✓				
$(\mathbb{Z}, -)$	✓				
$(\mathbb{Z}, \div)$	X				
$(\mathbb{Q}, +)$	✓				
$(\mathbb{Q}, \cdot)$	✓				
$(\mathbb{Q}, -)$	✓				
$(\mathbb{Q}, \div)$	$0 \div 0$ not defined X				
$(\mathbb{Q}^*, +)$	$(\frac{p}{q}) + (-\frac{p}{q}) = 0 \notin \mathbb{Q}^* \times$				
$(\mathbb{Q}^*, \cdot)$	✓				
$(\mathbb{Q}^*, -)$	$(\frac{p}{q}) - (\frac{p}{q}) = 0 \notin \mathbb{Q}^* \times$				
$(\mathbb{Q}^*, \div)$	✓				

Set of all non-zero rational no.





Topic : Semi-group

non-empty set

binary opⁿ

An algebraic structure $(S, *)$ is called a semi-group if and only if

$$(a * b) * c = a * (b * c), \quad \forall a, b, c \in S$$

i.e., binary opⁿ $*$ must be associative



	Algebraic Structure (Groupoid/Magma)	Semi-Group	Monoid	Group	Abelian Group
$(\mathbb{N}, +)$	✓	✓			
$(\mathbb{N}, \cdot)$	✓	✓			
$(\mathbb{N}, -)$	$2 - 3 = -1 \notin \mathbb{N} \times$	✗			
$(\mathbb{N}, \div)$	✗	✗			
$(\mathbb{Z}, +)$	✓	✓			
$(\mathbb{Z}, \cdot)$	✓	✓			
$(\mathbb{Z}, -)$	✓	✗ (Not associative)			
$(\mathbb{Z}, \div)$	✗	✗			
$(\mathbb{Q}, +)$	✓	✓			
$(\mathbb{Q}, \cdot)$	✓	✓			
$(\mathbb{Q}, -)$	✓	✗			
$(\mathbb{Q}, \div)$	$0 \div 0$ not defined ✗	✗			
$(\mathbb{Q}^*, +)$	$(\frac{p}{q}) + (-\frac{p}{q}) = 0 \notin \mathbb{Q}^* \times$	✗			
$(\mathbb{Q}^*, \cdot)$	✓	✓			
$(\mathbb{Q}^*, -)$	$(\frac{p}{q}) - (\frac{p}{q}) = 0 \notin \mathbb{Q}^* \times$	✗			
$(\mathbb{Q}^*, \div)$	✓	✗ (Not associative)			

Set of all non-zero rational no.
↓
 $\mathbb{Q}^*$



Topic : Monoid



A semi-group $(S, *)$ is called a monoid if and only if there exists an element $e \in S$ such that

$$a * e = e * a = a, \forall a \in S$$

'e' is called identity element of set 'S' w.r.t. binary opⁿ *



	Algebraic Structure (Groupoid/Magma)	Semi-Group	Monoid	Group	Abelian Group
$(\mathbb{N}, +)$	✓	✓	$0 \notin \mathbb{N}$ ✗		
$(\mathbb{N}, \cdot)$	✓	✓	$1 \in \mathbb{N}$ ✓		
$(\mathbb{N}, -)$	$2 - 3 = -1 \notin \mathbb{N}$ ✗	✗	✗		
$(\mathbb{N}, \div)$	✗	✗	✗		
$(\mathbb{Z}, +)$	✓	✓	$0 \in \mathbb{Z}$ ✓		
$(\mathbb{Z}, \cdot)$	✓	✓	$1 \in \mathbb{Z}$ ✓		
$(\mathbb{Z}, -)$	✓	✗ (Not associative)	✗		
$(\mathbb{Z}, \div)$	✗	✗	✗		
$(\mathbb{Q}, +)$	✓	✓	$0 \in \mathbb{Q}$ ✓		
$(\mathbb{Q}, \cdot)$	✓	✓	$1 \in \mathbb{Q}$ ✓		
$(\mathbb{Q}, -)$	✓	✗	✗		
$(\mathbb{Q}, \div)$	$0 \div 0$ not defined ✗	✗	✗		
$(\mathbb{Q}^*, +)$	$(\frac{p}{q}) + (-\frac{p}{q}) = 0 \notin \mathbb{Q}^*$ ✗	✗	✗		
$(\mathbb{Q}^*, \cdot)$	✓	✓	$1 \in \mathbb{Q}^*$ ✓		
$(\mathbb{Q}^*, -)$	$(\frac{p}{q}) - (\frac{p}{q}) = 0 \notin \mathbb{Q}^*$ ✗	✗	✗		
$(\mathbb{Q}^*, \div)$	✓	✗ (Not associative)	✗		

Set of all non-zero rational no.

$\mathbb{Q}^*$



Topic : Group

A monoid $(S, *)$ is called a group if and only if.

for each element $a \in S$,

there must exist some element $b \in S$

Such that $a * b = b * a = e$ (identity)

for a monoid to become a group for Element of set inverse must be present within the set itself

if $a * b = b * a = e$, then a and b are called inverse of each other w.r.t. binary opⁿ $*$



	Algebraic Structure (Groupoid/Magma)	Semi-Group	Monoid	Group	Abelian Group
(N, +)	✓	✓	$0 \notin N$ ✗	✗	{ inverse does not exist for any element Except for '1'
(N, ·)	✓	✓	$1 \in N$ ✓	✗	
(N, -)	$2 - 3 = -1 \notin N$ ✗	✗	✗	✗	
(N, ÷)	✗	✗	✗	✗	
(Z, +)	✓	✓	$0 \in Z$ ✓	$(a) + (-a) = 0$ (identity) ✓	{ inverse will not exist for any element Except for '1' and '-1' inv(1) = 1 inv(-1) = -1
(Z, ·)	✓	✓	$1 \in Z$ ✓	✗	
(Z, -)	✓	✗ (Not associative)	✗	✗	
(Z, ÷)	✗	✗	✗	✗	
(Q, +)	✓	✓	$0 \in Q$ ✓	✓	inv(0) will never exist w.r.t multiplication
(Q, ·)	✓	✓	$1 \in Q$ ✓	✗ $0 \in Q$	
(Q, -)	✓	✗	✗	✗	
(Q, ÷)	$0 \div 0$ not defined ✗	✗	✗	✗	
(Q*, +)	$(\frac{p}{q}) + (-\frac{p}{q}) = 0 \notin Q^*$ ✗	✗	✗	✗	
(Q*, ·)	✓	✓	$1 \in Q^*$ ✓	✓	
(Q*, -)	$(\frac{p}{q}) - (\frac{p}{q}) = 0 \notin Q^*$ ✗	✗	✗	✗	
(Q*, ÷)	✓	✗ (Not associative)	✗	✗	

Set of
all non-zero
rational no.

Q^*



Topic : Abelian group / Commutative group

A group $(S, *)$ is called an abelian (commutative) group. if and only if

$$a * b = b * a, \quad \forall a, b \in S$$

i.e. Commutative property must be satisfied by all the elements of the set

	Algebraic Structure (Groupoid/Magma)	Semi-Group	Monoid	Group	Abelian Group
$(\mathbb{N}, +)$	✓	✓	$0 \notin \mathbb{N}$ ✗	✗	✗
$(\mathbb{N}, \cdot)$	✓	✓	$1 \in \mathbb{N}$ ✓	✗	✗
$(\mathbb{N}, -)$	$2 - 3 = -1 \notin \mathbb{N}$ ✗	✗	✗	✗	✗
$(\mathbb{N}, \div)$	✗	✗	✗	✗	✗
$(\mathbb{Z}, +)$	✓	✓	$0 \in \mathbb{Z}$ ✓	✓	✓
$(\mathbb{Z}, \cdot)$	✓	✓	$1 \in \mathbb{Z}$ ✓	✗	✗
$(\mathbb{Z}, -)$	✓	✗ (Not associative)	✗	✗	✗
$(\mathbb{Z}, \div)$	✗	✗	✗	✗	✗
$(\mathbb{Q}, +)$	✓	✓	$0 \in \mathbb{Q}$ ✓	✓	✓
$(\mathbb{Q}, \cdot)$	✓	✓	$1 \in \mathbb{Q}$ ✓	✗	✗
$(\mathbb{Q}, -)$	✓	✗	✗	✗	✗
$(\mathbb{Q}, \div)$	$0 \div 0$ not defined ✗	✗	✗	✗	✗
$(\mathbb{Q}^*, +)$	$(\frac{p}{q}) + (-\frac{p}{q}) = 0 \notin \mathbb{Q}^*$ ✗	✗	✗	✗	✗
$(\mathbb{Q}^*, \cdot)$	✓	✓	$1 \in \mathbb{Q}^*$ ✓	✓	✓
$(\mathbb{Q}^*, -)$	$(\frac{p}{q}) - (\frac{p}{q}) = 0 \notin \mathbb{Q}^*$ ✗	✗	✗	✗	✗
$(\mathbb{Q}^*, \div)$	✓	✗ (Not associative)	✗	✗	✗

Set of all non-zero rational no.
↓
 $\mathbb{Q}^*$



Algebraic Structure	Semi-group	Monoid	Group
① Closed	① Closed ② '*' is associative	① Closed ② op ⁿ is associative ③ Identity element must be present within the set	① Closed ② '*' is associative ③ Identity element is present in the set ④ Inverse of Every element of the set must be present within the set itself



Topic : Note



① A set containing element '0' can never be a group w.r.t. multiplication because.

$\text{inv}(0) = \text{does not exist}$
w.r.t multiplication



Topic : Note



② $e * e = e$ (identity)
{Always true}

∴ Inverse of identity element is always the identity element itself.

③ Inverse of an element if exist in the set
w.r.t. binary opⁿ '*', then it is always unique

④ Inverse exist for Every element of the group
Within the group itself and it is unique



Topic : Note



$$(5) \quad (a * b)^{-1} = b^{-1} * a^{-1}$$

this will always hold true in a group
{ does not matter whether the group
is abelian or not }

(6) In a group,

$$(a^{-1})^{-1} = a$$

#Q. Let $A = \{0, \pm 2, \pm 4, \pm 6, \dots\}$
 $B = \{0 \pm 1, \pm 3, \pm 5, \dots\}$

Which of the following is not a semi-group

Semi-group $\begin{cases} \textcircled{1} \text{ Closed} \\ \textcircled{2} \text{ Associative} \end{cases}$

A

$(A, +)$

it is closed

it is associative

B

$(A, \cdot)$

it is closed

it is associative

C

$(B, +)$

it is associative

D

$(B, \cdot)$

it is closed

it is associative

$$1 + 3 = 4 \notin B$$

$\therefore B$ is not closed w.r.t. '+'

$\therefore$ Neither a groupoid, nor a semi-group

$\epsilon \in \{\text{ie., NULL string}\}$ is also included

#Q. Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings

ie. Σ^* is the set of all strings of 0 and 1 including ϵ (epsilon).

$$(10111 \oplus (101 \oplus 1101)) \oplus 1101 = 101111011101$$

$$(10111 \oplus 101) \oplus 1101 = 101111011101$$

$\therefore$ Associative

A

Not a semigroup

B

Semi group but not a monoid

$$(10 \oplus ?) = \epsilon$$

C

Monoid but not a group.

D

A group

① Closed ✓

② Associative ✓

③ Identity ✓

④ Inverse ✗

{Concatenation of any string 's' with ϵ will result the string 's' itself}

ie. ' ϵ ' is identity

inv(ϵ) = ϵ , but inverse does not exist for any other string of the set

H.W.



#Q. Let A be the set of all non-singular matrices of order $n \times n$ over real number and let $*$ be the matrix multiplication operation. Then

A A is closed under $*$ but $\langle A, * \rangle$ is not a semigroup

B $\langle A, * \rangle$ is a semigroup but not a monoid.

C $\langle A, * \rangle$ is a monoid but not a group.

D $\langle A, * \rangle$ is a group but not an abelian group.

H.W.

#Q. Let S be any finite set, and $F(S)$ is defined as set of all function on set S . Then $F(S)$ with respect to function composition operation (ie., $\circ$) is.

- A** Not a semigroup
- B** Semi group but not a monoid
- C** Monoid but not a group.
- D** A group

#Q. Let Z is the set of all integers. The binary operation $*$ is defined as $a*b = \max(a, b)$ then the structure $(Z, *)$ is

- A** Not a semigroup
- B** Semi group but not a monoid
- C** Monoid but not a group.
- D** A group

#Q. Let Q^+ be the set of all positive rational numbers. The binary operation $*$ is

defined as $a * b = \frac{ab}{3} \forall a, b, \in Q^+$ If $(Q^+, *)$ is a group then find

- (i) identity element of the group
- (ii) inverse of any element $a, \forall, \in \text{Group}$

#Q. Which of the following statement is/are not true.

- A** $\{0, \pm 2k, \pm 4k, \pm 6k, \dots\}$ is a group with respect to addition where k is any fixed positive integer
- B** $\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication
- C** $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication
- D** None of these



2 mins Summary



Topic

Algebraic structure

Topic

Semi-group

Topic

Monoid

Topic

Group

Topic

Abelian group / Commutative group

Topic

Practice questions

Slide



THANK - YOU